PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Chirp – A Real-Time Social Media Interaction Platform

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Chirp – A Real-Time Social Media Interaction Platform"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Shikha Tuteja**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Chirp is a modern full-stack social media web application designed to enable users to share posts, interact with other users, and communicate in real time through a seamless and responsive interface. For developing the application, the React.js library is employed in the frontend process for the designing of a strong user interface that includes many interactive functionalities. The Node.js and Express.js technologies will be used by us for the development of our application in the backend process. Thus, we will be able to create an effective API for handling all user interaction-related requests.

In terms of database interactions, we plan to use Prisma ORM library. It allows you to perform safe and type-safe interactions with databases. Password hashing and secure token-based authentication are implemented in the application. In addition, the application includes functionality provided by Socket.io that allows implementing live messaging, notifications, and user interactions without reloading the page. As a result, the application uses the potential of modern web design and communication technologies, delivering a scalable social networking platform.

1. Introduction

1.1 Background and Context

Social media websites have been playing an important part in communications over the Internet. Users today are expecting software solutions which are fast, safe, scalable and interactive in nature. Chirp has been created with the purpose of showcasing how modern-day web technologies can be used to create an interactive social media website.

1.2 Problem Statement

Social media platforms may encounter problems associated with scalability, real-time messaging, and authentications of users. Ineffective designs could make the platform slow and insecure. Chirp resolves these concerns by offering a full stack solution, which is equipped with real-time capabilities and effective user management.

1.3 Significance of the Project

This project is an example of practical implementation of advanced web designing techniques as well as Full Stack development approach. The project shows practical implementation of technologies associated with front end, back end, database administration, and real time communications within one software application.

1.4 Objectives

- Create a full stack social networking site.
- Use user authentication for the application.
- Integrate real-time interaction between users.
- Build a sustainable application.

1.5 Scope and Limitations

Scope:

- User Authentication and Posting Management

- Real-time Communication Features
- Easy-to-use Web Interface

Limitations:

- Limited moderation capabilities
- Lacks multi-language capabilities

2. Methodology

The design process of the Chirp application is based on the Agile methodology. It allows implementing iterative development, continuous testing, and improvement of functionality. Agile methodology facilitates the development of the project in incremental cycles, identifying possible issues and further improving it according to the functional needs of the product. The separate development and optimization of each feature ensured its high performance.

Modular development was used to design each feature separately. This includes authentication, post management, and real-time interaction of users with one another. Modular development increases the maintenance capacity of the application and makes debugging much more accessible. Furthermore, it can improve future scalability and feature addition.

2.1 Technical Approach

This application uses a full-stack technology stack appropriate for scalable and real-time web applications.

- The React.js library is applied in frontend development for creating a dynamic user interface.
- Node.js environment and Express.js framework are used for backend development.
- Prisma ORM library is used to manage database operations.
- Socket.io library is used to make real-time application possible.

It is assured that the selected technology stack will enable effective communication between the frontend and backend parts while providing high performance and scalability.

2.2 Workflow Description

Application workflow consists of several steps.

1. Registration and login of users.
2. Authentication using tokens and password encryption.
3. Managing posts by creating and interacting through likes and comments.
4. Instant interactions are performed using Socket.io.
5. All data of users are stored securely in the database managed with Prisma.
6. Updated content is retrieved from the database using RESTful APIs by the frontend.

2.3 System Architecture

Application architecture involves client-side and server-side elements as follows:

- The client-side aspect is covered by React.js in the development of user interfaces.
- Node.js will be applied as the backend, wherein Express.js will be used for APIs and other aspects of business logic.
- Prisma ORM will be applied for managing database and ensuring data protection.
- Socket.io will be employed for establishing real-time communication among users. Data will be secured through a combination of token authentication and encrypted passwords.

3. Tools and Technologies

3.1 Programming Languages

- JavaScript (Node.js + React.js)
- HTML
- CSS

3.2 Frameworks / Libraries

- React.js (Frontend)
- Express.js (Backend)
- Tailwind CSS (UI Styling)
- JWT (JSON Web Token) for Authentication
- Prisma ORM (Database Interaction)
- Socket.io (Real-Time Communication)

3.3 Databases / Storage

- Relational Database managed using **Prisma ORM**
- Secure and scalable data storage

3.4 IDEs / Platforms

- Visual Studio Code
- GitHub (Version Control)

3.5 APIs / External Integrations

- RESTful APIs for frontend-backend communication
- Socket.io for real-time events and messaging

4. Implementation

During the implementation phase, the design is turned into a fully functioning social media application. The development of the Chirp application is done using a modern full-stack architecture which allows for scalability, modularity, and security. Every essential feature is implemented in a separate module making future improvements possible.

4.1 Authentication Module

The Authentication Module makes sure that access to the Chirp platform is provided in a safe way. The users have to be registered first and then login using their correct credentials. When the login is successfully completed, JSON Web Tokens (JWT) are generated and stored. They are added to the headers of protected API requests, and the middleware validates each request in order to authenticate it. Therefore, this module offers a secure session management and protects private user data.

4.2 Feedback Submission Module

Users can create, read and manipulate posts via the Feedback Submission Module. They can submit new posts, view posts created by other users and communicate with them through likes or other interactions.

Both client-side and server-side input validations are performed in order to make sure about the quality of data. All the posts are saved into the database together with all needed metadata like timestamps and interactions.

4.3 Admin Dashboard Module

The Real-Time Communication Module is built based on Socket.io framework. It makes real-time communication between users without reloading pages possible.

Instantly happening events improve the performance of the application and offer a much better user experience.

4.4 Database Design

The project uses structured database schema managed by Prisma ORM library. The models include users and posts as well as their properties. User data like authentication credentials and profile data are stored in the database.

5. Major Findings/Outcomes/Output/Results

Several important conclusions have been drawn after the creation, testing, and evaluation of the Chirp Social Media Application. All those conclusions highlight the ability to combine modern full-stack technologies to create secure, reliable, and user-oriented web applications.

5.1 Secure and Scalable Social Media Platform

The Chirp application has proved to offer a platform that is highly secure and scalable. JWT-based authentication secures access to resources, and secure token verification, as well as middleware-based authorization, guarantees data integrity and access management. Moreover, the scalable architecture of the backend part of the application makes it possible to add new features and services and still provide efficient performance.

5.2 Improved User Engagement through Real-Time Interaction

Use of real-time communication technologies implemented by means of Socket.io helps to improve engagement and interaction. There is no need for refreshing a website to get updated on the changes. As a consequence, server polling is minimized, which contributes to enhanced performance of the system. Thus, user engagement is significantly increased.

5.3 Enhanced Security through Role-Based Access Control

A more secure system is achieved due to the implementation of the principle of role-based access control (RBAC). Each user has access only to those actions allowed by their role.

Sensitive actions can be performed only with authorization. Thus, higher security is achieved due to an organized access control management within the application.

5.4 Reliable and Efficient Data Management

With the use of Prisma ORM, efficient and consistent database management can be achieved. Type-safe database operations provided by the ORM allow for avoiding runtime errors. High concurrency and data transaction management capabilities are provided. Therefore, the application is capable of high performance under load conditions.

5.5 Collaborative and Maintainable Development using GitHub

The version control system based on GitHub provided effective collaboration among developers during the process. Such functionalities as branching and pull requests made the work easier. Code readability was improved as well as its maintenance and debugging.

5.6 Clean and Modular Application Architecture

Another important achievement of the project can be highlighted as an efficient clean and modular architecture implementation. It is provided by the separation of the frontend, the backend, and the database. It results in high readability, maintainability, and independent development and testing of modules.

6. Conclusion and Future Scope

6.1 Conclusion

The Chirp social media application is a perfect illustration of creating a modern full stack web platform by adding a responsive front end, secure back end, correct database management, and communication features.
This project provides a valuable example of best practices in software development. It proves that full-stack technologies can be used to develop efficient real-life applications.

6.2 Future Scope / Enhancements

Despite the fact that the existing implementation covers all basic functional requirements, there is still a number of improvements that can be made to enhance the functionality of Chirp even more.

6.2.1 AI-Based Content Recommendation

AI and Machine Learning technologies can be employed to analyze user activity. As a result, the platform will be able to suggest personalized content to increase user interaction and improve retention.

6.2.2 Advanced Notification System

An advanced notification system can be developed. Notifications will be delivered instantly whenever there is any action on the profile and messages of the users.

6.2.3 Analytics and User Insights

Various advanced analytics tools may be introduced for gathering data regarding users' interaction with the website. These insights can be used by managers to make appropriate decisions.

6.2.4 Mobile Application Development

Creating a mobile app compatible with Android and iOS platforms is another improvement that needs to be considered. Making the Chirp website accessible through your mobile phone will increase its usability.

6.2.5 Content Moderation and Safety Tools

Automatic moderation tools are a necessity on every website in order to detect all violations and create a safe environment.

7. References

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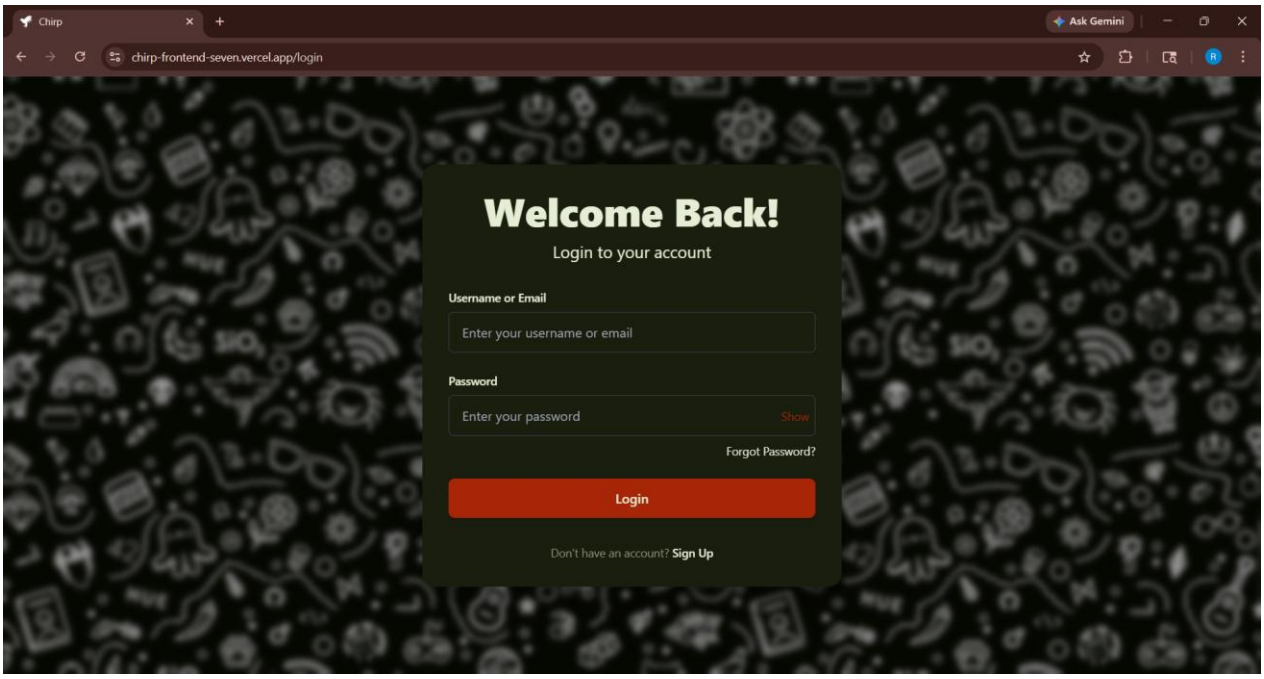
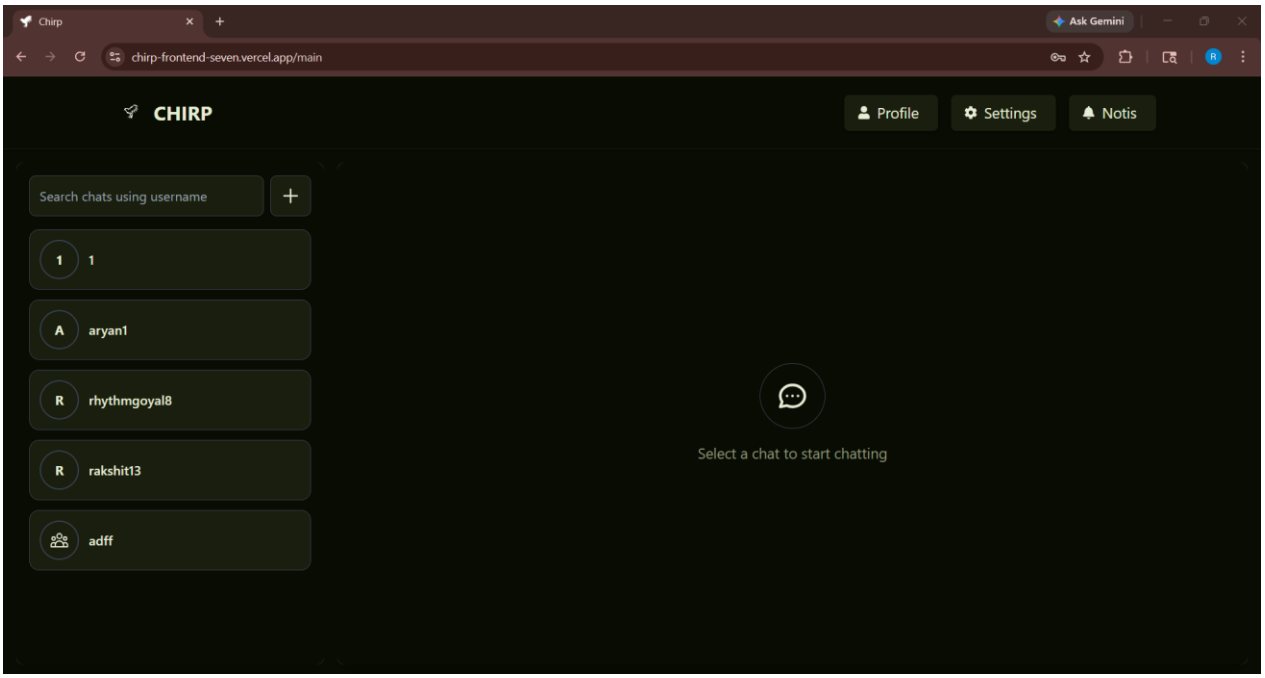
8. Appendices

Screenshots of project

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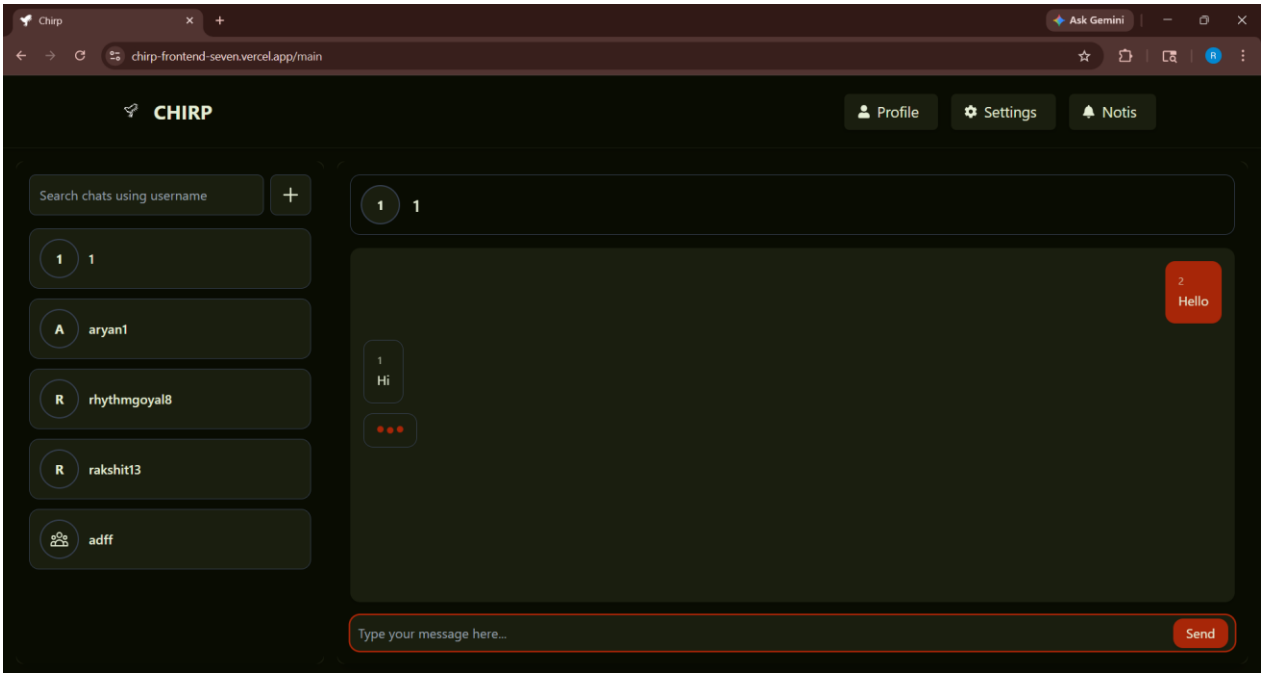
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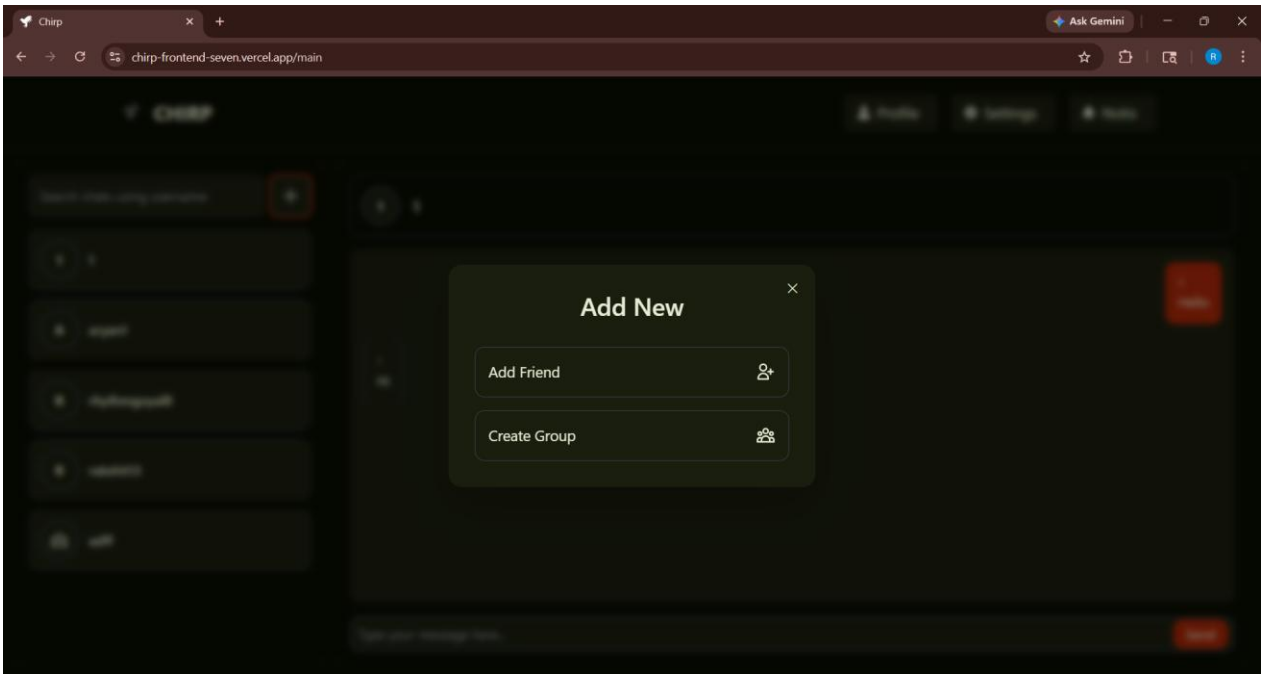

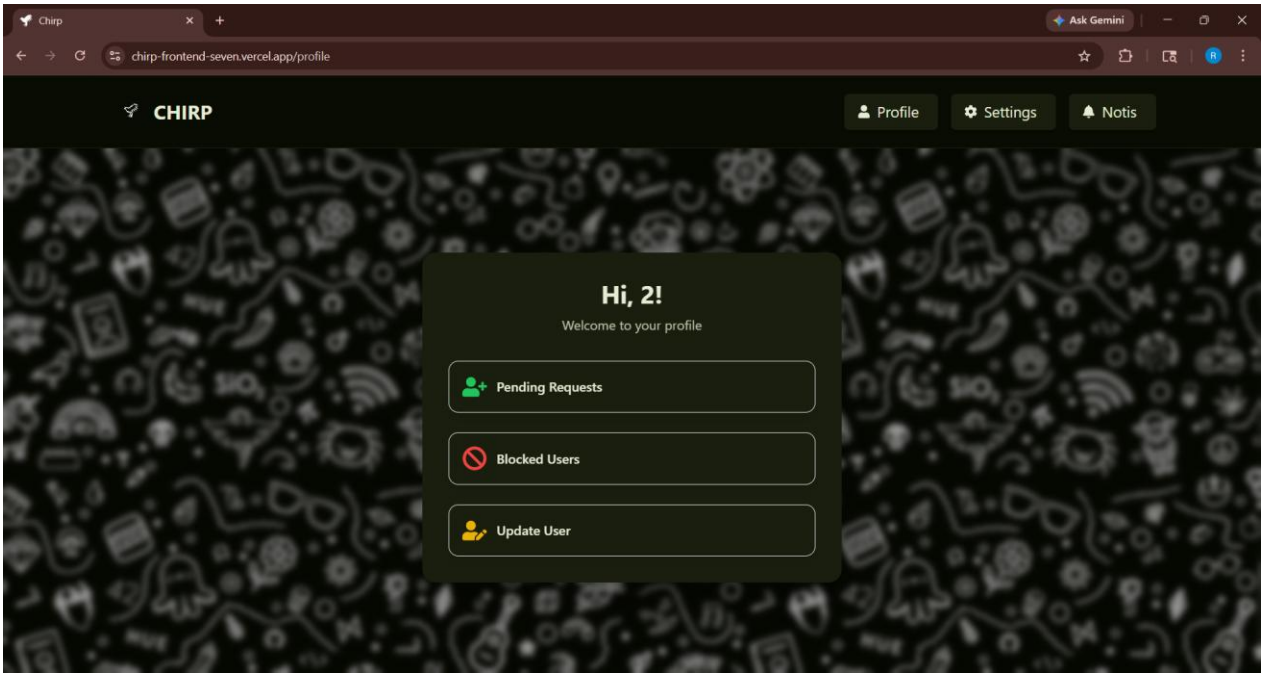

Source code available at https://github.com/rishabhdevKaushik/Chirp-A-Real-Time-Social-Media-Interaction-Platform-2210992156_2210992134_2210992147_2210991418